DEPARMENT OF INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY
UNIVERSITY OF SINDH, JAMSHORO

IT25-425 P-II
COMPUTER NETWORKS
MORNING/EVENING

LAB#4

TASK: CREATING NEW TOPOLOGIES WITH PACKET TRACER

Name: _____ Roll No: _____

Score: _____ Signature of the Lab Tutor _____ Date: _____

Objectives:

1. **To discuss and understand different topologies in networking.**

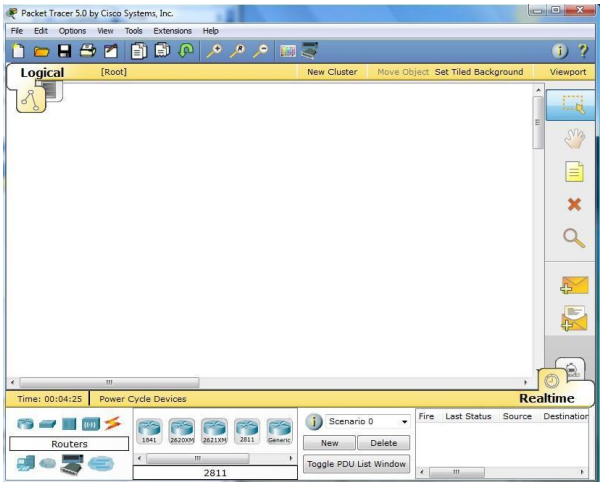
Equipment:

- PC with operating system
- Packet Tracer 8.2 or latest

Discussion:

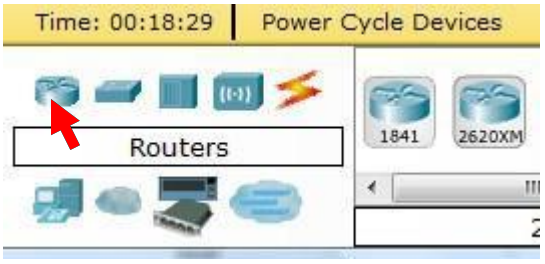
Packet Tracer is a protocol simulator developed by Dennis Frezzo and his team at Cisco Systems. Packet Tracer (PT) is a powerful and dynamic tool that displays the various protocols used in networking, in either Real Time or Simulation mode. This includes layer 2 protocols such as Ethernet and PPP, layer 3 protocols such as IP, ICMP, and ARP, and layer 4 protocols such as TCP and UDP. Routing protocols can also be traced.

Step 1: Start Packet Tracer

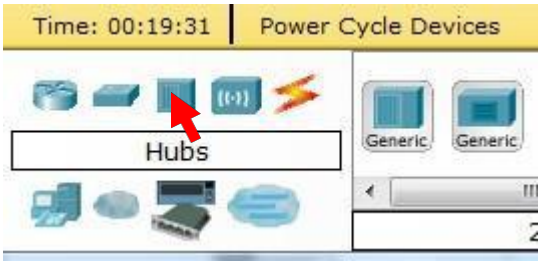


Step 2: Choosing Devices and Connections

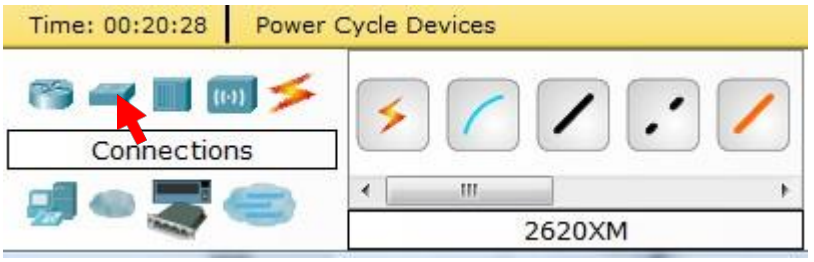
We will begin building our network topology by selecting devices and the media in which to connect them. Several types of devices and network connections can be used. For this lab we will keep it simple by using End Devices, Switches, Hubs, and Connections. Single click on each group of devices and connections to display the various choices. The devices you see may differ slightly.







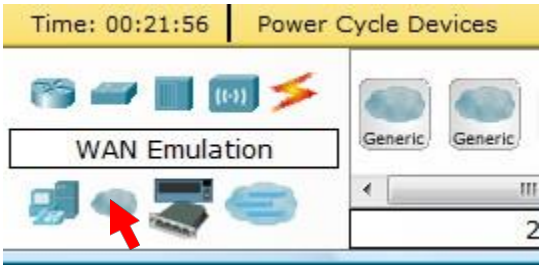










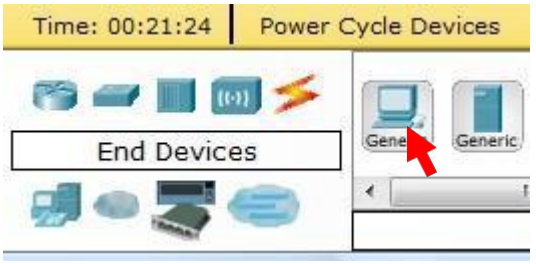


Step 3: Building the Topology – Adding Hosts

Single click on the **End Devices**.

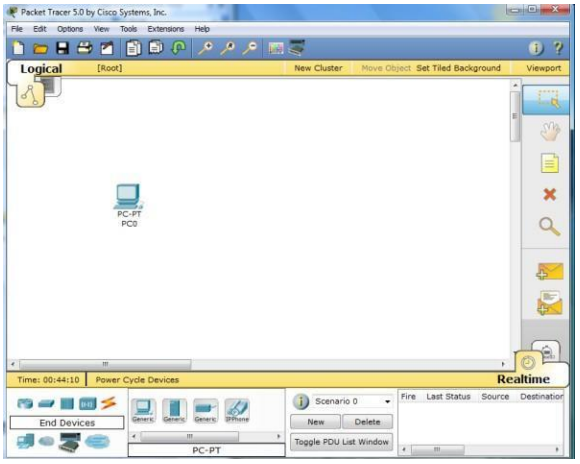


Single click on the **Generic** host.



Move the cursor into topology area. You will notice it turns into a plus "+" sign.

Single click in the topology area and it copies the device.



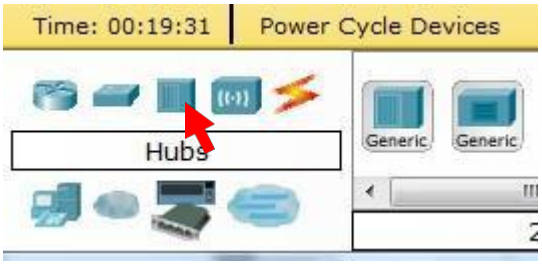
Add three more hosts.

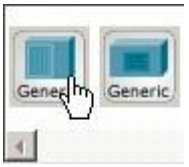


Step 4: Building the Topology – Connecting the Hosts to Hubs and Switches

Adding a Hub

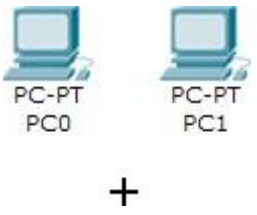
Select a hub, by clicking once on Hubs and once on a Generic hub.

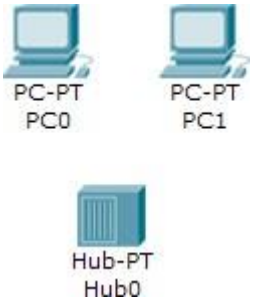




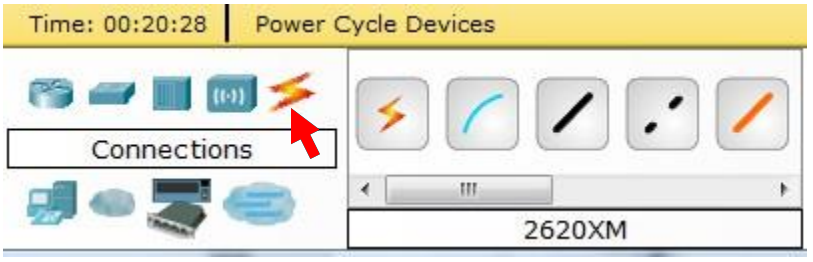


Add the hub by moving the plus sign "+" below PC0 and PC1 and click once.

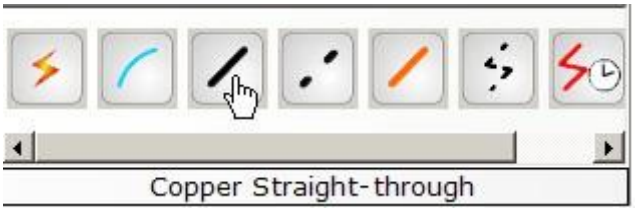




Connect PC0 to Hub0 by first choosing Connections.



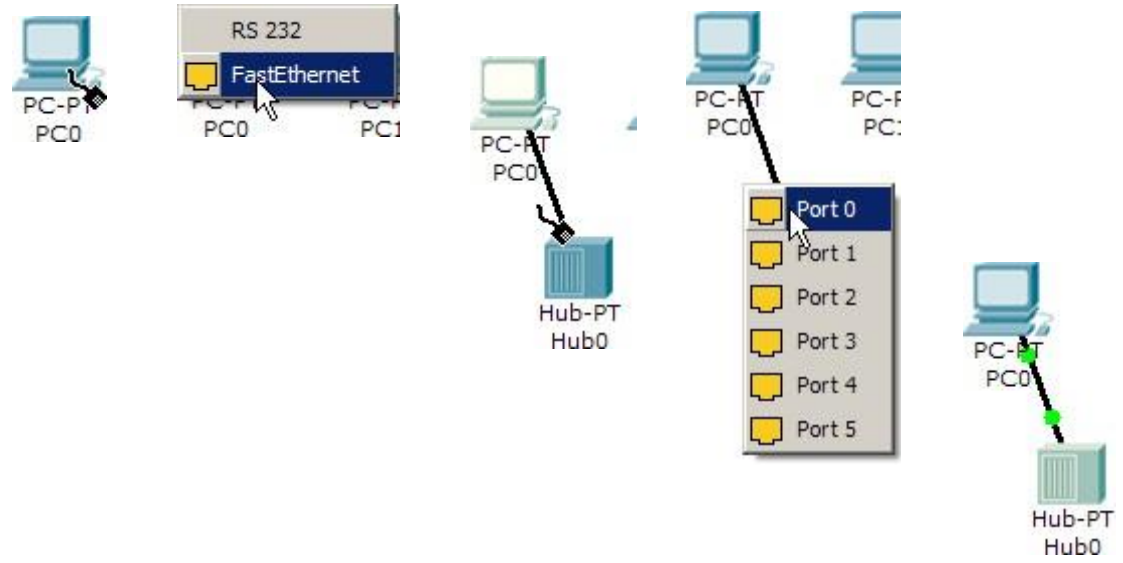
Click once on the Copper Straight-through cable.



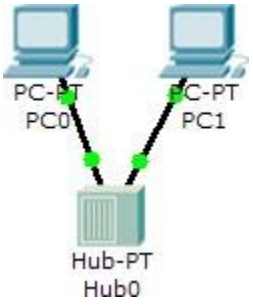
Perform the following steps to connect PC0 to Hub0:

Click once on PC0, Choose FastEthernet, Drag the cursor to Hub0
Click once on Hub0 and choose Port 0
Notice the green link lights on both the PC0 Ethernet NIC and the Hub0 Port 0 showing that the link is active.

1 2 3 4 5

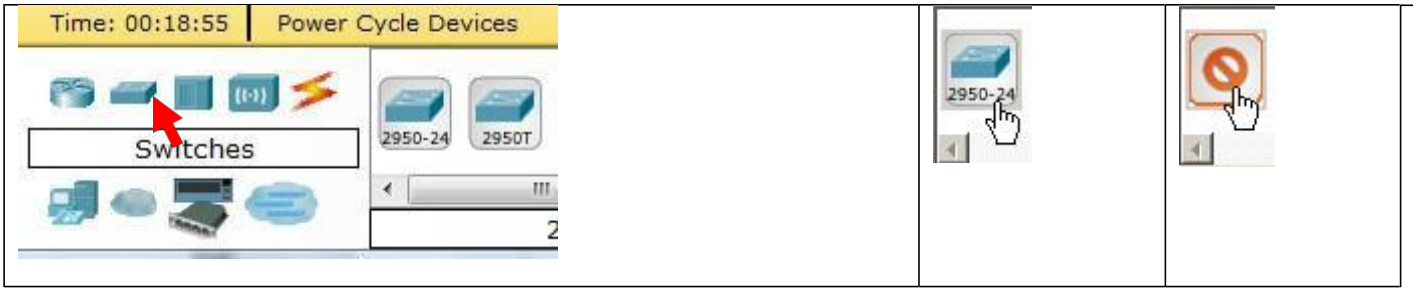


Repeat the steps above for PC1 connecting it to Port 1 on Hub0. (The actual hub port you choose does not matter.)

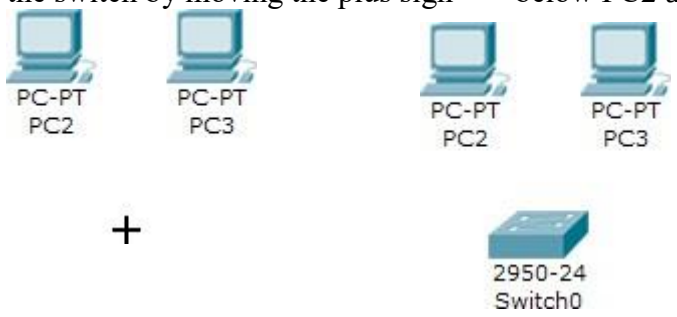


Adding a Switch

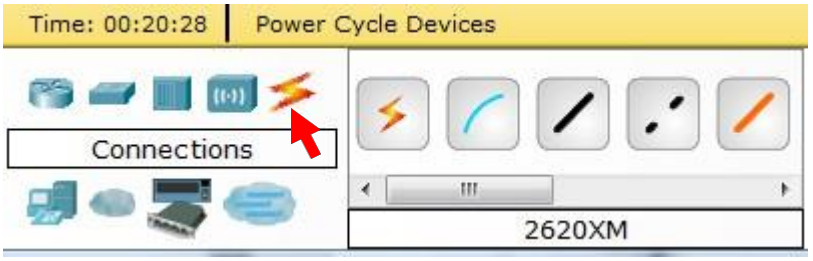
Select a switch, by clicking once on Switches and once on a 2950-24 switch.



Add the switch by moving the plus sign “+” below PC2 and PC3 and click once.



Connect PC2 to Hub0 by first choosing Connections.



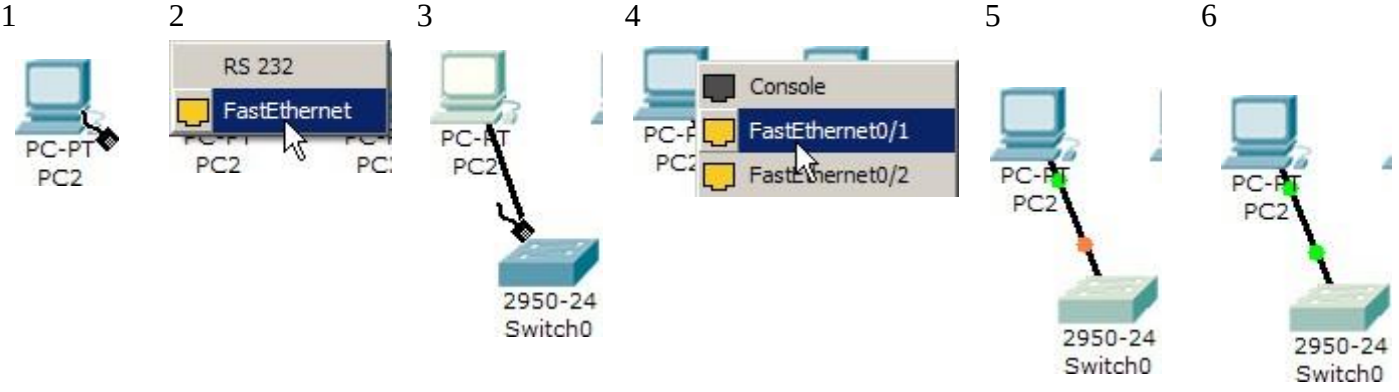
Click once on the Copper Straight-through cable.



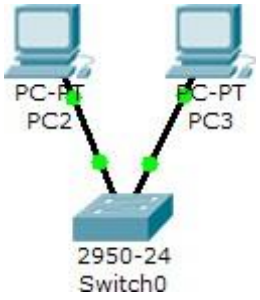
Perform the following steps to connect PC2 to Switch0:
Click once on PC2 Choose FastEthernet
Drag the cursor to Switch0
Click once on Switch0 and choose FastEthernet0/1
Notice the green link lights on PC2 Ethernet NIC and amber light Switch0 FastEthernet0/1 port. The switch port is temporarily not forwarding frames, while it goes through the stages for the Spanning Tree Protocol (STP) process.
After a about 30 seconds the amber light will change to green indicating that the port has entered the forwarding stage. Frames can now forwarded out the switch port.

Note: Spanning Tree Protocol (STP) is discussed in Theory Classes.

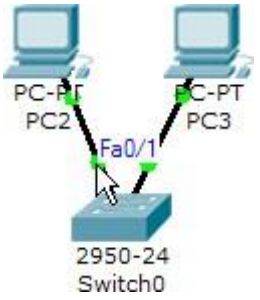
1 2 3 4 5 6



Repeat the steps above for PC3 connecting it to Port 3 on Switch0 on port FastEtherent0/2. (The actual switch port you choose does not matter.)



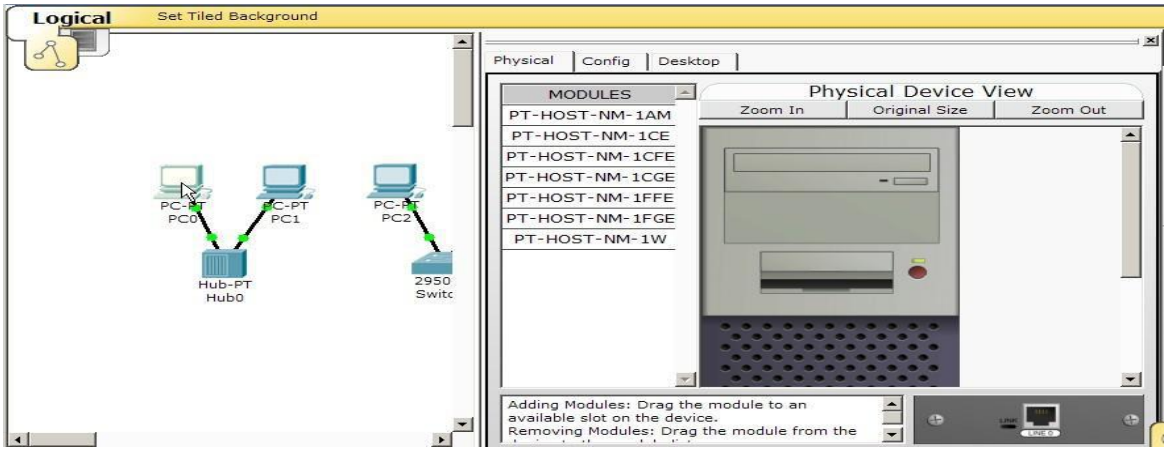
Move the cursor over the link light to view the port number. Fa means FastEthernet, 100 Mbps Ethernet.



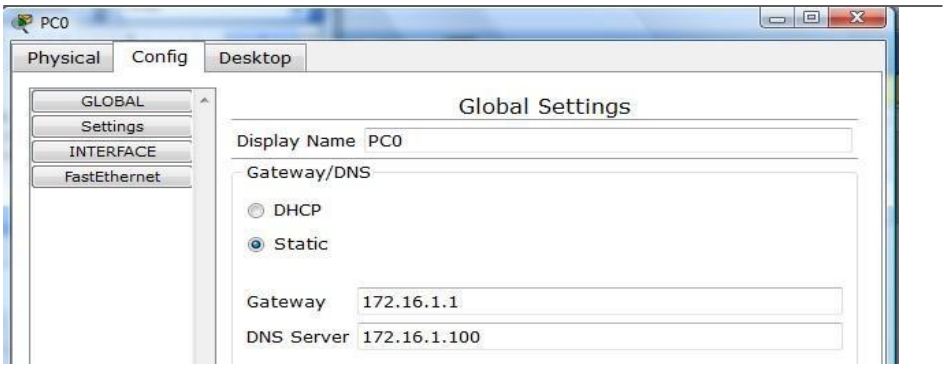
Step 5: Configuring IP Addresses and Subnet Masks on the Hosts

Before we can communicate between the hosts we need to configure IP Addresses and Subnet Masks on the devices.

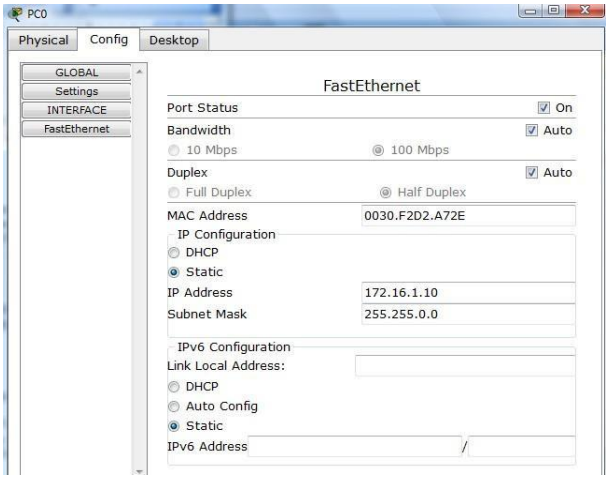
Click once on PC0.



Choose the Config tab and click on Settings. It is here that you can change the name of PC0. It is also here where you would enter a Gateway IP Address, also known as the default gateway and the DNS Server IP Address. We will discuss this later, but this would be the IP address of the local router. If you want, you can enter the Gateway IP Address 172.16.1.1 and DNS Server IP Address 172.16.1.100, although it will not be used in this lab.



Click on Interface and then FastEthernet. Although we have not yet discussed IP Addresses, add the IP Address to 172.16.1.10. Click once in the Subnet Mask field to enter the default Subnet Mask. You can leave this at 255.255.0.0. We will discuss this later.



Also, notice this is where you can change the Bandwidth (speed) and Duplex of the Ethernet NIC (Network Interface Card). The default is Auto (autonegotiation), which means the NIC will negotiate with the hub or switch. The bandwidth and/or duplex can be manually set by removing the check from the Auto box and

choosing the specific option.

Bandwidth -
Auto
If the host is connected to a hub or switch port which can do 100 Mbps, then the Ethernet NIC on the host will choose 100 Mbps (Fast Ethernet). Otherwise, if the hub or switch port can only do 10 Mbps, then the Ethernet NIC on the host will choose 10 Mbps (Ethernet).

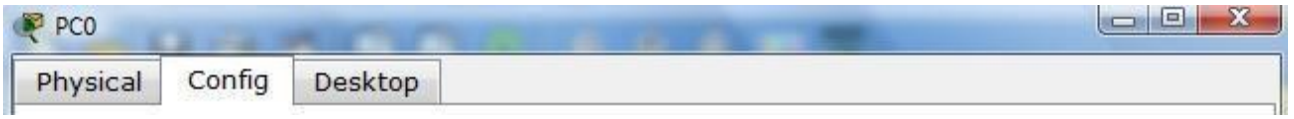
Duplex -
Auto

Hub: If the host is connected to a hub, then the Ethernet NIC on the host will choose Half Duplex.

Switch: If the host is connected to a switch, and the switch port is configured as Full Duplex (or Autonegotiation), then the Ethernet NIC on the host will choose Full Duplex. If the switch port is configured as Half Duplex, then the Ethernet NIC on the host will choose Half Duplex. (Full Duplex is a much more efficient option.)

The information is automatically saved when entered.

To close this dialog box, click the “X” in the upper right.

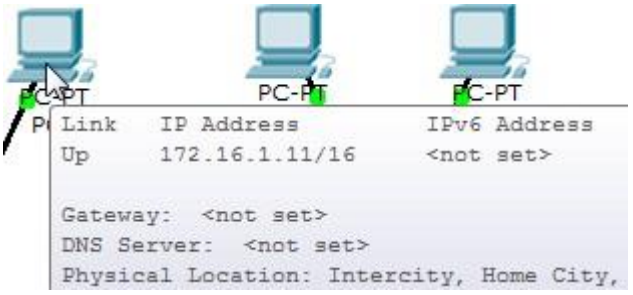


Repeat these steps for the other hosts. Use the information below for IP Addresses and Subnet Masks.

Host	IP Address	Subnet Mask
PC0	172.16.1.10	255.255.0.0
PC1	172.16.1.11	255.255.0.0
PC2	172.16.1.12	255.255.0.0
PC3	172.16.1.13	255.255.0.0

Verify the information:

To verify the information that you entered, move the Select tool (arrow) over each host.

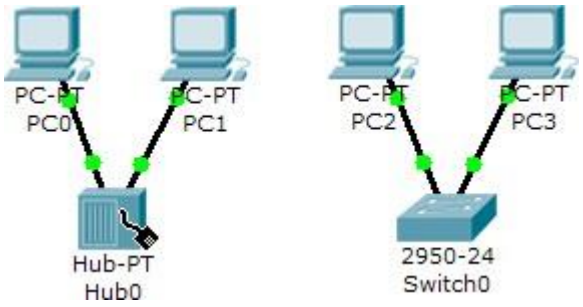


Deleting a Device or Link:

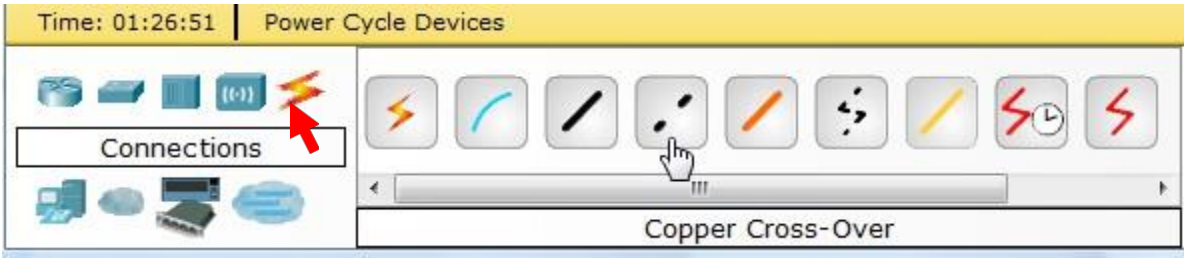
To delete a device or link, choose the Delete tool and click on the item you wish to delete.



Step 6: Connecting Hub0 to Switch0

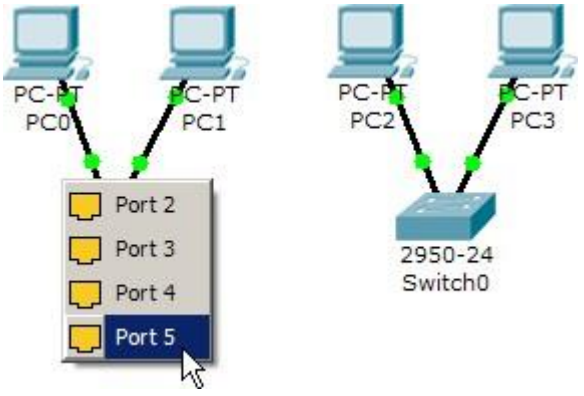


To connect like-devices, like a Hub and a Switch, we will use a Cross-over cable. Click once the Cross-over Cable from the Connections options.

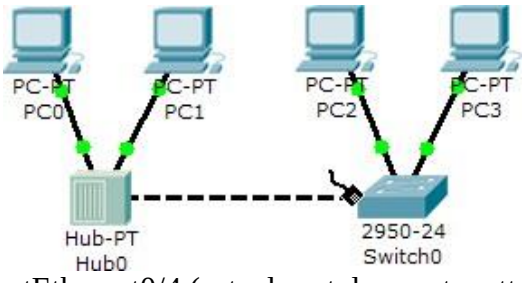


Move the Connections cursor over Hub0 and click once.

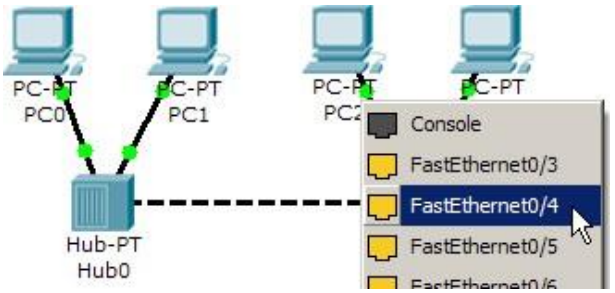
Select Port 5 (actual port does not matter).



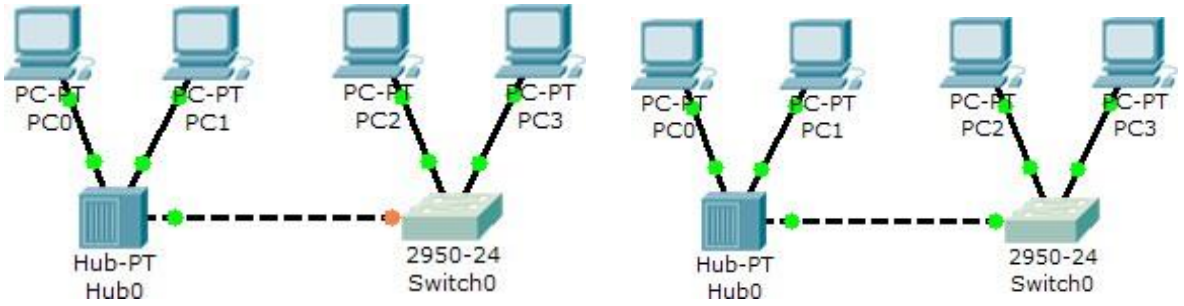
Move the Connections cursor to Switch0.



Click once on Switch0 and choose FastEthernet0/4 (actual port does not matter).



The link light for switch port FastEthernet0/4 will begin as amber and eventually change to green as the Spanning Tree Protocol transitions the port to forwarding.



Exercise:

Question 1: When choosing devices and connections, what do mean by Generic devices? Answer:

_____

_____

Question 2: Which Class of IP Address is used in the following practical? Answer:

_____

_____

Question 3: What is the different between Ethernet, Fast Ethernet and Port on Hub? Answer:

_____

_____

Question 4: While connecting switch and hub, which connection will be required and why? Answer:

_____

_____

Note: Submit the handout before the next Lab